

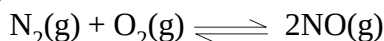
WORK SHEET - 02

THERMODYNAMICS

Single correct

- Q.1 A definite amount of an ideal gas ($\gamma = 1.5$) undergoes a change of state in which heat exchange is equal to work done ($q = w$). Molar heat capacity of the gas is :
(A) $2R$ (B) $3R$ (C) R (D) Infinite
- Q.2 When two equal sized pieces of the same metal at different temperatures T_h (hot piece) and T_c (cold piece) are brought into contact into thermal contact and isolated from its surrounding. The total change in entropy of system is given by
(A) $C \ln \frac{T_c + T_h}{2T_c}$ (B) $C \ln \frac{T_2}{T_1}$ (C) $C \ln \frac{(T_c + T_h)^2}{2T_h \cdot T_c}$ (D) $C \ln \frac{(T_c + T_h)^2}{4T_h \cdot T_c}$
- Q.3 Which of the options contain correct set of parameters which are both amount and path independent parameters.
(I) Internal energy (II) Specific enthalpy
(III) Heat exchange at constant pressure condition
(IV) Molar heat exchange at constant volume condition
(V) Temperature (VI) Work involved
(A) Only II, IV & V (B) All option except VI
(C) All options except III, IV & VI (D) Only II & V
- Q.4 A real gas is subjected to an adiabatic process causing a change in state from (3 bar, 50 L, 500K) to (5 bar, 40 L, 600 K) against a constant pressure of 4 bar. The magnitude of enthalpy change for the process is :
(A) 4000J (B) 5000J (C) 9000J (D) 1000J
- Q.5 Pressure of 10 moles of an ideal gas is changed from 2 atm to 1 atm against constant external pressure without change in temperature. If surrounding temperature (300 K) and pressure (1 atm) always remains constant then calculate total entropy change ($\Delta S_{\text{system}} + \Delta S_{\text{surrounding}}$) for given process.
[Given : $\ln 2 = 0.70$ and $R = 8.0 \text{ J/mol/K}$]
(A) 56 J/K (B) 14 J/K (C) 16 J/K (D) None of these
- Q.6 A system contains 1 mole of a monoatomic ideal gas. Now 1 mole of a diatomic nonreacting ideal gas is added into the system at constant volume and temperature. Due to addition of diatomic gas, choose the **incorrect** statement, regarding the new system.
(A) Enthalpy of system will increase
(B) Adiabatic coefficient ($\gamma = C_p/C_v$) of the system will decrease.
(C) Internal energy of the system remains constant.
(D) Pressure of the system will increase.
- Q.7 The enthalpy of tetramerization of X in gas phase ($4X(g) \rightarrow X_4(g)$) is -100 kJ/mol at 300 K. The enthalpy of vaporisation for liquid X and X_4 are respectively 30 kJ/mol and 72 kJ/mol respectively. ΔS for tetramerization of X in liquid phase is -125 J/K mol at 300 K. What is the ΔG at 300 K for tetramerization of X in liquid phase ?
(A) -52 kJ/mol (B) -89.5 kJ/mol (C) -14.5 kJ/mol (D) None of these
- Q.8 Which of the reaction is expected to be spontaneous at low temperatures and non-spontaneous at high temperatures.
(A) Dissociation of PCl_5 to give PCl_3 and Cl_2 (B) Formation of H_2O from $\text{H}_2(g)$ and $\text{O}_2(g)$
(C) Melting of ice (D) Vaporisation of any volatile liquid

Q.9 Calculate ΔG (kJ/mole) for the reaction at 300K.



at an instant where partial pressure of N_2 , O_2 and NO are 10^{-1} bar, 10^{-3} bar, 10^{-3} bar. If $\Delta H_f^\circ \text{NO}(\text{g})$ at 300K = 90.5 kJ/mole and $\Delta S_f^\circ \text{NO}(\text{g})$ at 300K = 12.5 J/Kmole [$2.303 \times R \times 300 = 5750 \text{ J/mole}$]

(A) 173.5 kJ/mole (B) 185 kJ/mole (C) 162 kJ/mole (D) 84.25 kJ/mole

Q.10 Select the option which is correct regarding the following statements.

Statement-1 : Entropy of system always increases in a spontaneous process.

Statement-2 : Adiabatic process whether reversible or irreversible will not cause any change in entropy of the surroundings.

Statement-3 : CO molecule will have residual entropy at 0 K.

(A) All the statements are true.

(B) Only statement-1 is incorrect.

(C) Only statement-3 is correct.

(D) Only statement-2 is correct.

Q.11 Which of the following options contain correct set of cases where **entropy of system is increasing**?

Case-I Dissociation of ammonia gas into N_2 and H_2 gases.

Case-II Polymerization process of ethene to form polythene.

Case-III Adiabatic irreversible compression of an ideal gas

Case-IV Condensation of $\text{H}_2\text{O}_{(\text{vap})}$ to give $\text{H}_2\text{O}_{(\text{l})}$

Case-V Isothermal free expansion of Ideal gas.

(A) All the cases

(B) Case-I, III and V only

(C) Case-I, II and V only

(D) Case-I, II, III and V only

Q.12 A substance has latent heat of vaporisation (at its boiling point 300 K) = 3 kJ/gm. If molar mass of substance is 40, then molar entropy change for condensation process will be :

(A) 10 J/K

(B) 400 J/K

(C) - 400 J/K

(D) - 10 J/K

Q.13 The heat capacity of a diatomic ideal gas subjected to a process such that absolute temperature is proportional to cube of its volume is given by :

(A) $\frac{5R}{2}$

(B) $\frac{7R}{2}$

(C) $\frac{17R}{6}$

(D) $\frac{3R}{2}$

Assertion Reason

Q.14 **Statement-1 :** Adiabatic free expansion of any substance in a closed system will not cause any change in temperature of the substance.

Statement-2 : In adiabatic free expansion of any substance in a closed system internal energy will remain constant.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.

Assertion Reason

Q.15 **Statement-1 :** Adiabatic free expansion of any substance in a closed system will not cause any change in temperature of the substance.

Statement-2 : In adiabatic free expansion of any substance in a closed system internal energy will remain constant.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.

More than one correct :

Q.16 Which of the following process must be spontaneous?

(A) Isobaric and Isoentropic process with negative enthalpy change.

(B) Isochoric and Isothermal process with negative internal energy change.

(C) A process with constant internal energy and Isothermal with an increase in entropy of system.

(D) An adiabatic process with constant volume condition and entropy of system increasing.

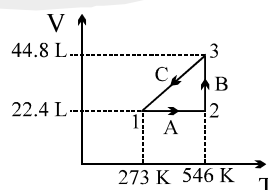


- Q.17 Which of the following option(s) will show a decrease in Gibbs free energy?
 (A) Combustion of propane at 1 bar and 500K.
 (B) Vapourisation of any liquid at 1 atm and above its normal boiling point.
 (C) Fusion of H_2O at 1 atm and $0^\circ C$ if its normal melting point is $0^\circ C$.
 (D) Vapourisation of H_2O at $100^\circ C$ and 1 bar if its normal boiling point is $100^\circ C$.
- Q.18 Which of the following statements is / are incorrect.
 (A) Combustion of methane in an adiabatic rigid container will cause no change in temperature of the system.
 (B) It is possible to have both adiabatic reversible & adiabatic irreversible processes between two states.
 (C) For a reaction involving only ideal gases, occurring at constant temperature there will be no change in internal energy.
 (D) P–V work is always non zero when there is change in volume.
- Q.19 Assume ideal gas behaviour for all the gases considered & vibrational degrees of freedom to be active. Separate equimolar samples of Ar, H_2 , SO_2 were subjected to a two step process as mentioned. Initially all are at same state of temperature & pressure.
Step I $\longrightarrow$ All undergo reversible adiabatic expansion to attain same final volume, which is double the original volume thereby causing the decreases in their temperature.
Step II $\longrightarrow$ After step I all are given appropriate amount of heat isochorically to restore the original temperature.
 Mark the correct option (s).
 (A) Due to step I only, the decrease in temperature will be maximum for Ar
 (B) During step II, heat given will be maximum for SO_2
 (C) There will be no net change in internal energy for any of the gas after both the steps of process are completed.
 (D) The P–V graph of Ar & H_2 will be different for overall process.

Subjective

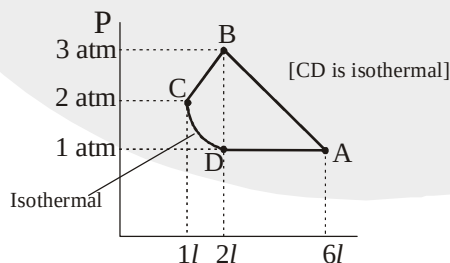
- Q.20 0.5 mole each of two ideal gases A ($C_v = \frac{3}{2} R$) and B ($C_v = \frac{5}{2} R$) are taken in a container and expanded reversibly and adiabatically from $V = 1$ L to $V = 4$ L starting from initial temperature $T = 300$ K. Find $|\Delta H|$ for the process in (Cal/mol).
- Q.21 A system containing an ideal gas ($\gamma = 1.5$) at 10 bar, 1 L and 300K undergoes expansion upto 4L. Calculate heat exchange of the system (in J) if process is carried out at a constant external pressure of 1 bar and final pressure becomes equal to 5 bar.
- Q.22 One mole of an ideal monoatomic gas is carried through the reversible cycle of the given figure consisting of step A, B and C and involving state 1,2 and 3. Fill in the blank space in the table given below assuming reversible steps.

State	P	V	T
1			
2			
3			



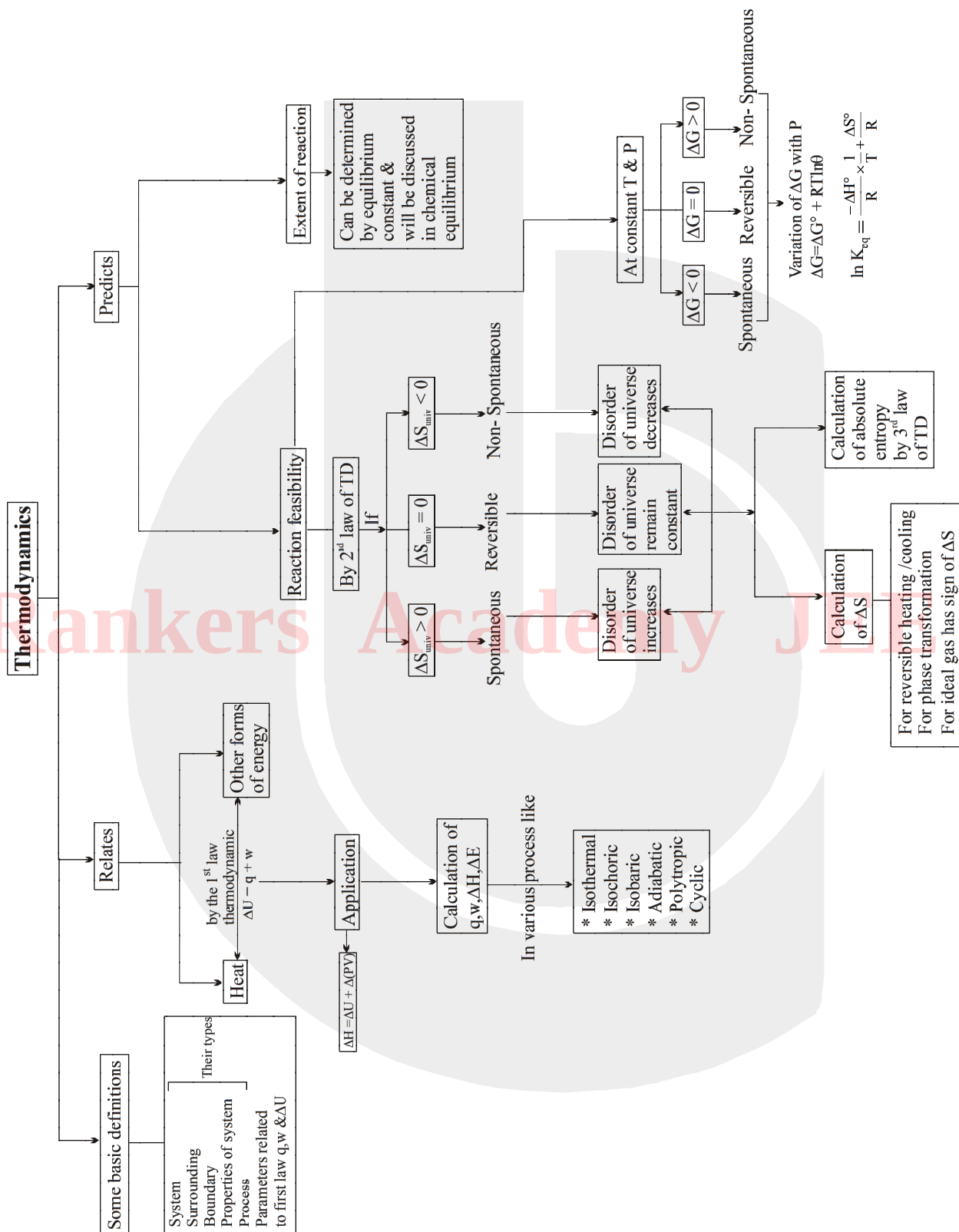
Step	Name of process	q	w	ΔE	ΔH
A					
B					
C					

- Q.23 When 1 mole of ice melt at 0°C and at constant pressure of 1 atm. 1440 calories of heat are absorbed by the system. The molar volumes of ice and water are 0.0196 and 0.0180 litre respectively. Calculate ΔH and ΔU for the reaction.
- Q.24 The molar enthalpy of vaporization of benzene at its boiling point (353 K) is $30.84 \text{ kJ mol}^{-1}$. What is the molar internal energy change? For how long would a 12 volt source need to supply a 0.5 A current in order to vaporise 7.8 g of the sample at its boiling point ?
- Q.25 20.0 dm^3 of an ideal gas (diatomic $C_{v,m} = 5R/2$) at 673 K and 0.7 MPa expands until pressure of the gas is 0.2 MPa. Calculate $q, w, \Delta U$ and ΔH for the process if the expansion is: [Given: $(7/2)^{5/7} = 2.447$]
 (i) Isothermal and reversible
 (ii) Adiabatic and reversible.
 (iii) Isothermal and adiabatic
 (iv) Against 0.2 MPa and adiabatic
 (v) Against 0.2 MPa and isothermal.
- Q.26 Container A containing one mole of He is connected with Container B containing 2 moles of Ne at the same temperature and pressure. Calculate ΔS_{mixing} . [Given: $R = 2 \text{ Cal K}^{-1} \text{ mol}^{-1}$; $\ln 3 = 1$; $\ln 2 = 0.70$]
[Report your answer by multiplying it with 100]
- Q.27 The equilibrium constant of the reaction $2\text{C}_3\text{H}_6(\text{g}) \rightleftharpoons \text{C}_2\text{H}_4(\text{g}) + \text{C}_4\text{H}_8(\text{g})$ is found to fit the expression
- $$\ln K_{\text{eq}} = -1.04 - \frac{1088}{T/\text{K}}$$
- (where T/K is temperature expressed in Kelvin scale)
 Calculate the standard reaction enthalpy and entropy at 400 K.
- Q.28 A substance undergoes a change represented as shown $A_{(l)} [1 \text{ bar}, 300 \text{ K}] \longrightarrow A_{(s)} [1 \text{ bar}, 300 \text{ K}]$. From the given information, calculate magnitude of change in internal energy (in Joules) when 1 moles of $A(l)$ solidifies.
- Information :**
- Standard melting point of $A(s)$ is 300 K.
 - Latent heat of fusion of A is 0.01 kJ/gm .
 - Specific volume of $A(s)$ is 100 ml/g and $A(l)$ is 120 ml/gm .
 - Molar mass of A is 50 gm / mole .
- Q.29 One mole of a monoatomic ideal gas is subjected to a process as shown. Calculate the maximum temperature obtained during the process (in Kelvin)



[If $T_{\text{max}} = \frac{X}{R}$, then fill X in the OMR sheet.]

REVISION FLOW CHART



LIST OF IMPORTANT FORMULAS

Heat Capacity : $C = \frac{\theta}{\Delta T} \Rightarrow C_{p,m} - C_{v,m} = R \text{ and } r = \frac{C_{p,m}}{C_{v,m}}$

$$C_{v,m} = \left(\frac{f_t}{2} + \frac{f_R}{2} + f_{vib} \right) \times R \quad (f \rightarrow \text{degree of freedom})$$

$$dH = n \int C_{p,m} dT \Rightarrow dU = n \int C_{v,m} dT \Rightarrow W = - \int P_{ext} dV$$

Enthalpy Calculation :

$$\Delta H = \Delta U + \Delta(PV) = \Delta U + (P_2 V_2 - P_1 V_1)$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$(W_{iso})_{reversible} = -nRT \ln \frac{V_2}{V_1} = -nRT \ln \frac{P_1}{P_2}$$

$$(W_{iso})_{reversible} = -P_{ext}(V_2 - V_1) = -P_{ext} \left[\frac{nRT}{P_2} - \frac{nRT}{P_1} \right]$$

$$W_{adiabatic} = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} = nC_{v,m} dT = \frac{nR}{\gamma - 1} (T_2 - T_1)$$

$$\left. \begin{array}{l} PV^\gamma = \text{constant} \\ TV^{\gamma-1} = \text{constant} \\ P^{\gamma-1} T^\gamma = \text{constant} \end{array} \right\} \text{ applicable for reversible process only}$$

For Polytropic process :

$$W = \frac{P_2 V_2 - P_1 V_1}{x - 1} = \frac{nR(T_2 - T_1)}{x - 1} \Rightarrow C_m = C_{v,m} + \frac{R}{1 - x}$$

Second Law of Thermodynamics :

$$\text{Efficiency of Carnot Cycle} = 1 - \frac{T_L}{T_H}$$

Calculation of Entropy Change :

(a) **For reversible heating / Cooling of liquids / solids**

$$\Delta S = nC_m \ln \frac{T_f}{T_i}$$

(b) **For Real Gas :** $\Delta S = nC_{p,m} \ln \frac{T_f}{T_i}$ (isobaric) $\Rightarrow \Delta S = nC_{v,m} \ln \frac{T_f}{T_i}$ (isochoric)

(c) **For Phase Transformation:** $\Delta_{vap} S = \frac{\Delta_{vap} H}{T_{BP}}$ $\Delta_{fus} S = \frac{\Delta_{fus} H}{T_{MP}}$

(d) **For Ideal Gas undergoing any process**

$$\Delta S = nC_{v,m} \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i}$$

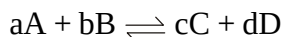
$$\Delta S = nC_{p,m} \ln \frac{T_f}{T_i} + nR \ln \frac{V_i}{V_f}$$

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surrounding}}$$

Third Law of Thermodynamics :

$$\Delta S_m^0 \text{ (TK)} = \int_0^{\text{mp}} \frac{C_{p,m} dT}{T} + \frac{\Delta_{\text{fus}} H}{T_{\text{mp}}} + \int_{\text{mp}}^{\text{BP}} \frac{C_{p,m}' dT}{T} + \frac{\Delta_{\text{vap}} H}{T_{\text{BP}}} + \int_{\text{BP}}^{\text{TK}} \frac{C_{p,m}'' dT}{T}$$

(a) For Chemical Reaction :



$$\Delta_{\text{reaction}} S = \sum (x_i S_i)_{\text{product}} - \sum (x_i S_i)_{\text{reactant}}$$

(b) Variation of ΔS with Temperature:

$$\Delta S_{T_2} = \Delta S_{T_1} + \Delta_{\text{reaction}} C_p \ln \left(\frac{T_f}{T_i} \right)$$

(c) Variation of ΔS with Pressure

$$\Delta S_{P_2} = \Delta S_{P_1} + \Delta n_g R \ln \left(\frac{P_1}{P_2} \right)$$

Variation of Enthalpy with Temperature :

$$\Delta H_{T_2} = \Delta H_{T_1} + \int_{T_1}^{T_2} \Delta_{\text{rxn}} C_p dT$$

$$\Delta U_{T_2} = \Delta U_{T_1} + \int_{T_1}^{T_2} \Delta_{\text{rxn}} C_v dT$$

Gibbs Free Energy :

$$U = H - TS$$

$$\Delta G = -T \Delta S_{\text{total}}$$

$$\Delta G = \Delta H - T \Delta S \quad (\text{at constant } P \text{ and } T)$$

$$\Delta G = VdP - SdT$$

Variation of ΔG with P at constant temperature ($dT = 0$) :

$$\Delta G = nRT \ln \frac{P_2}{P_1} \quad (\text{For ideal gas})$$

$$\Delta G = V(P_2 - P_1) \quad (\text{For solids and liquids})$$

$$\Delta G = \Delta G^\circ + RT \ln \theta$$

$$\ln K_{\text{eq}} = -\frac{\Delta H^\circ}{R} \times \frac{1}{T} + \frac{\Delta S^\circ}{R}$$

$$\ln K_{\text{eq}} = -\frac{\Delta H^\circ}{R} \times \frac{1}{T} + \frac{\Delta S^\circ}{R} \quad (\text{Its graph})$$

$\downarrow$
Y

$\downarrow$
m

$\downarrow$
x

$\downarrow$
C

$$\log \frac{K_2}{K_1} = \frac{\Delta H^\circ}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

ANSWER KEY

Q.1	C	Q.2	D	Q.3	A	Q.4	C	Q.5	C
Q.6	C	Q.7	C	Q.8	B	Q.9	C	Q.10	B
Q.11	B	Q.12	C	Q.13	C	Q.14	D	Q.15	D
Q.16	AD	Q.17	ABD	Q.18	ABCD	Q.19	ABCD		
Q.20	[− 900 calories]			Q.21	[2300]				

Q.22

State	P	V	T
1	1 atm	22.4	273
2	2 atm	22.4	546
3	1 atm	44.8	546

Step	Name of process	q	w	ΔE	ΔH
A	Isochoric	$3/2 R(273)$	0	$3/2 R(273)$	$5/2 R(273)$
B	Isothermal	$546 R \ln 2$	$-546 R \ln 2$	0	0
C	Isobaric	$-5/2 R(273)$	$R(273)$	$-3/2 R(273)$	$-5/2 R(273)$

- Q.23 [1440 calories]
 Q.24 $\Delta U = 27.91 \text{ kJ mol}^{-1}$, $t = 514 \text{ sec}$.
 Q.25 (i) $q = -w = 17.54 \text{ kJ}$, $\Delta U = 0$ and $\Delta H = 0$;
 (ii) $q = 0$, $w = \Delta U = -10.536 \text{ kJ}$ and $\Delta H = -14.75 \text{ kJ}$
 (iii) $q = 0$, $w = 0$, $\Delta U = 0$ and $\Delta H = 0$
 (iv) $q = 0$; $\Delta U = w = -7.14 \text{ kJ}$; $\Delta H = -9.996 \text{ kJ}$,
 (v) $q = -w = 10.0 \text{ kJ}$, $\Delta U = \Delta H = 0$
 Q.26 [320]
 Q.27 $\Delta H^\circ = 9.04 \text{ kJ/mol}$; $\Delta S^\circ = -8.64 \text{ J/mol}^{-1}\text{K}^{-1}$
 Q.28 [0400]
 Q.29 [0008]